

Project Report

SupplyPrescript Analytics

Closed-Loop Prescriptive Analytics for Supply Chain Operations

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Project Title: Supply Prescriptive Analytics

Internship Project: Axlero Solutions

1. Project Objective

The objective of SupplyPrescript is to build a closed-loop prescriptive analytics platform for supply chain operations. The system predicts shipment delays using machine learning, recommends corrective actions using mathematical optimization, supports operational decision execution, records actual outcomes and decision ROI, and uses feedback to continuously improve the prediction model.

2. Overall System Workflow

Shipment Data	XGBoost Delay Prediction	Optimization & Recommendations
Decision Execution	Feedback & Actual Outcome	ROI Evaluation
Discrepancy Detection	Model Retraining	Continuous Learning

3. Week 1 – Predictive Analytics

In Week 1, I developed a predictive analytics baseline using an XGBoost Regressor to estimate shipment delay in days. A synthetic supply chain dataset was created using supplier reliability, shipment distance, weather risk, and historical lead time. The model was trained and evaluated by comparing predicted delays with actual delay values.

Features used: Supplier Rating, Distance (miles), Weather Risk Index, Historical Lead Time.

The Week 1 implementation established the predictive foundation used by the later optimization and closed-loop components.

4. Week 2 – Mathematical Optimization & Prescriptive Analytics

In Week 2, a prescriptive analytics layer was added using mathematical optimization and SciPy Linear Programming. When a delay is predicted, the optimization engine evaluates possible corrective actions under operational constraints and returns multiple strategies.

- **Low Cost Strategy:** Lower cost with slower delivery.
- **Fast Delivery Strategy:** Faster delivery with higher operational cost.
- **Balanced Strategy:** A compromise between cost and delivery speed.

Each recommendation can include selected actions, total cost, time impact, and a cost-versus-speed trade-off explanation.

5. Week 3 – Closed-Loop Feedback & Decision ROI

In Week 3, the project was extended from prediction and recommendation into a closed-loop evaluation system. The system records actual outcomes after a decision is executed and compares them with the predicted outcome. Feedback is stored for evaluation and future improvement.

- Compare predicted outcomes with actual business outcomes.
- Record feedback from executed decisions.
- Track decision effectiveness and Decision ROI.
- Provide an analytics view for reviewing decision outcomes.

6. Week 4 – Continuous Learning & Model Retraining

In Week 4, a Continuous Learning pipeline was implemented. The pipeline identifies discrepancies between predicted and actual outcomes. When the prediction error meets the defined retraining condition, the XGBoost model can be retrained using updated feedback and historical outcome data. The updated model is then validated for future predictions.

This completes the closed-loop cycle: the system does not stop after making a prediction or recommendation; it uses real-world outcomes to improve future predictions.

7. SupplyPrescript Analytics Dashboard

The interactive dashboard connects the complete workflow, including delay prediction, optimization, strategy recommendations, decision execution, feedback, ROI analytics, and model retraining.

Dashboard evidence included in the GitHub repository:

- Main Dashboard
- Delay Prediction
- Optimization
- Strategy Recommendations
- Decision Execution
- Feedback & Closed-Loop Evaluation
- Analytics & Decision ROI
- Continuous Learning & Retraining

8. Tools & Technologies Used

Category	Technology
Programming	Python
Machine Learning	XGBoost, scikit-learn
Optimization	SciPy Linear Programming
Data Processing	pandas, NumPy
API / Backend	FastAPI
Dashboard	Retool
Development	Jupyter Notebook
Model Persistence	Joblib

9. Final Project Outcome

SupplyPrescript demonstrates how predictive analytics, prescriptive optimization, operational execution, feedback evaluation, and continuous learning can be combined into a single supply chain decision-support workflow. The project moves beyond simply predicting what may happen by helping an analyst determine what action should be taken, evaluate the result, and use the outcome to improve future decisions.

10. Conclusion

This project helped me understand how predictive and prescriptive analytics work together in supply chain management. The XGBoost model predicts shipment delays, mathematical optimization generates alternative corrective strategies, the dashboard supports decision execution and feedback, ROI analytics evaluates outcomes, and the continuous learning pipeline enables model improvement. Together, these components form a closed-loop analytics system designed to support more informed and effective supply chain decisions.